SpaceX Falcon 9 Landing Prediction using Data Science

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EXECUTIVE SUMMARY

- This project analyzes SpaceX Falcon 9 launch data to identify factors that influence the successful landing of the first stage booster. By performing exploratory data analysis on features such as payload, launch site, and orbit, meaningful patterns are identified. These insights can be used to build predictive models that estimate landing success, helping improve mission efficiency and reduce costs through booster reusability.

INTRODUCTION

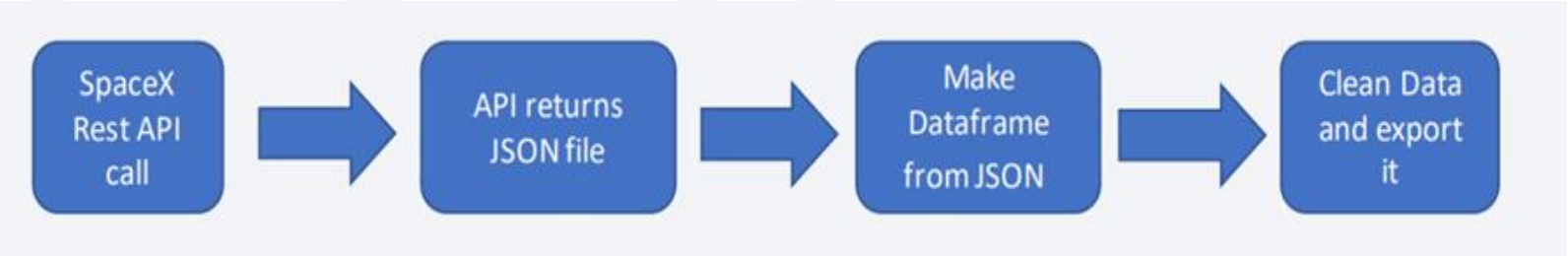
- SpaceX aims to reduce the cost of space missions through reusable Falcon 9 boosters.
- Successful first-stage landing plays a key role in achieving reusability.
- The project analyzes historical launch data to understand landing outcomes.
- Key factors such as payload, orbit, and launch site are studied.
- Data analysis and machine learning techniques are used to predict landing success.

METHODOLOGY

- Data collected using SpaceX REST API and Wikipedia web scraping.
- Data wrangling performed (cleaning, removing unnecessary columns, encoding)
- Exploratory Data Analysis (EDA) using visualization and SQL queries
- Interactive analysis using Folium maps and Plotly Dash dashboards
- Predictive analysis using machine learning classification models
- Models built, tuned, and evaluated to select the best-performing model

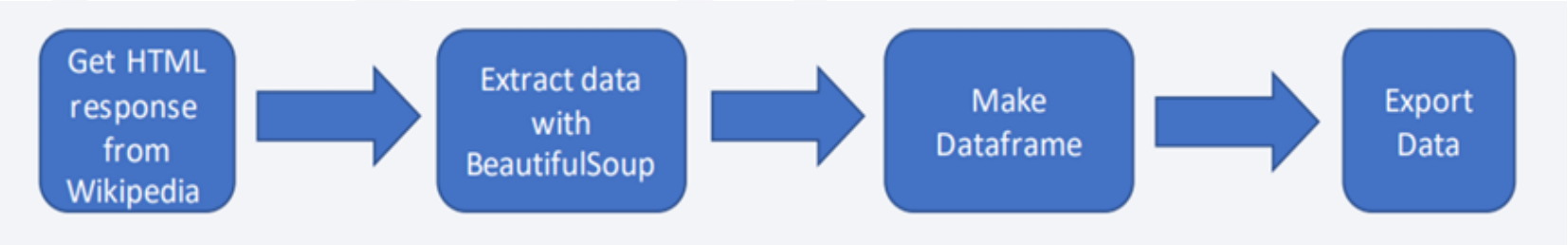
Data Collection – Space API

- Data is collected using the **SpaceX REST API**
- API endpoint used: api.spacexdata.com/v4/
- The API provides:
  - Launch details
  - Rocket information
  - Payload data
- Steps involved:
  - Send request to API and receive response
  - Convert response into JSON format
  - Extract relevant fields (launch, payload, booster data)
  - Transform data into a structured dataframe
  - Clean and filter the dataset
  - Export data for further analysis



Data Collection – Web Scraping

- Data is collected by scraping **Wikipedia launch records**
- Source URL contains Falcon 9 and Falcon Heavy launch data
- The scraped data includes:
  - Launch details
  - Landing outcomes
  - Payload information
- Steps involved:
  - Send HTTP request to Wikipedia page
  - Parse HTML content using BeautifulSoup
  - Extract required tables and columns
  - Clean and structure the extracted data
  - Convert data into a dataframe
  - Export dataset for further analysis



Data Wrangling

- Dataset contains both successful and unsuccessful booster landings
- Landing outcomes include:
  - True (Ocean, RTLS, ASDS) → Success
  - False (Ocean, RTLS, ASDS) → Failure
- Convert categorical outcomes into numerical format:
  - 1 → Successful landing
  - 0 → Failed landing
- Create a new column “Landing Class” for classification
- Clean and preprocess data for further analysis

EDA with data visualisation

- Used visualization techniques to explore relationships in launch data
- Scatter Plots:
  - Analyze correlation between variables (e.g., Flight Number, Payload, Orbit, Launch Site)
- Bar Chart:
  - Compare success rate across different orbit types
- Line Graph:
  - Show trend of launch success over years
- Helps in identifying patterns, correlations, and trends in the dataset

EDA with SQL

- Used SQL queries to extract insights from the dataset
- Identify unique launch sites
- Filter records based on conditions (e.g., launch site, year)
- Analyze payload statistics:
    - Total payload mass
    - Average payload mass
- Determine landing outcomes:
    - First successful landing date
    - Count of successful and failed missions
- Identify:
    - Boosters with maximum payload
    - Successful landings under specific conditions

Build an Interactive Map with Folium

- Folium map object is a map centered on NASA Johnson Space Center at Houson, Texas
- Red circle at NASA Johnson Space Center's coordinate with label showing its name (folium.Circle, folium.map.Marker).
- Red circles at each launch site coordinates with label showing launch site name (folium.Circle, folium.map.Marker, folium.features.DivIcon).
- The grouping of points in a cluster to display multiple and different in formation for the same coordinates (folium.plugins.MarkerCluster).
- Markers to show successful and unsuccessful landings. Greenfor successful landing and Redfor unsuccessful landing. (folium.map.Marker, folium.Icon).
- Markers to show distance between launch site to key locations (railway, highway, coastway, city) and plot a line between them. (folium.map.Marker, folium.PolyLine, folium.features.DivIcon)
- These objects are created in order to understand better the problem and the data. We can show easily all launch sites, their surroundings and the number of successful and unsuccessful landings.

Build a Dashboard with PlotlyDash

- Developed an interactive dashboard using Plotly Dash
- Dropdown Menu:
  - Select launch site or view all sites
- Pie Chart:
  - Shows success vs failure for selected site
- Range Slider:
  - Filter data based on payload mass
- Scatter Plot:
  - Displays relationship between Payload Mass and Landing Outcome
- Enables interactive analysis and better data understanding

Predictive Analysis (Classification)

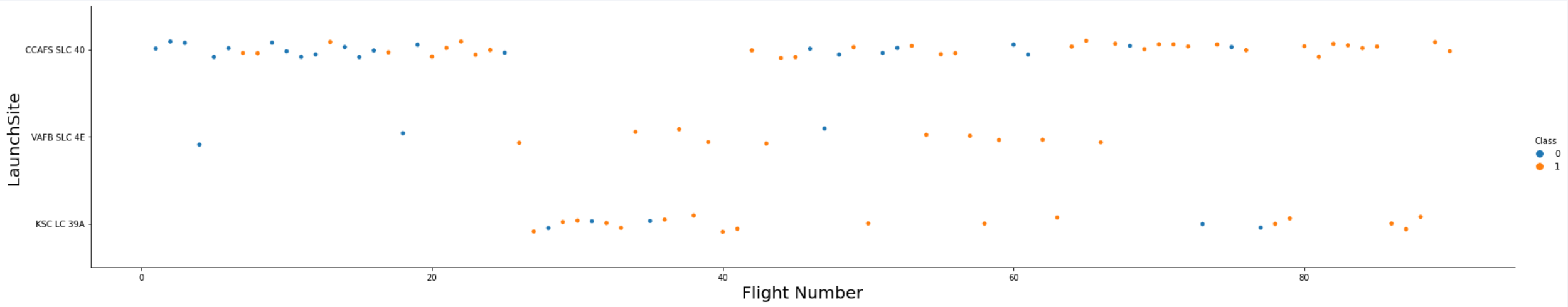
- **Data Preparation:**
  - Load and normalize dataset
  - Split data into training and testing sets
- **Model Building:**
  - Apply classification algorithms
  - Tune hyperparameters using GridSearchCV
- **Model Evaluation:**
  - Evaluate models using test accuracy
  - Analyze performance using confusion matrix
- **Model Comparison:**
  - Compare different models based on accuracy
  - Select the best-performing model

Results

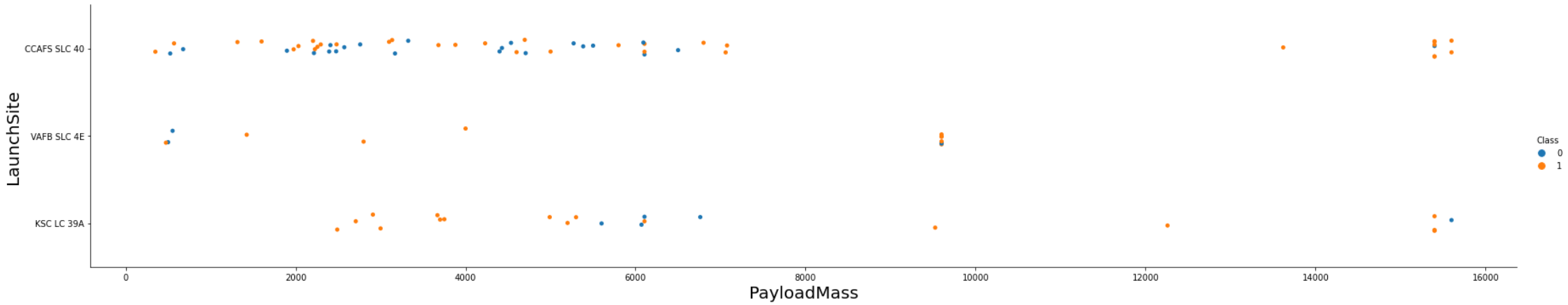
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Insights Drawn From EDA

Flight Number vs. LaunchSite

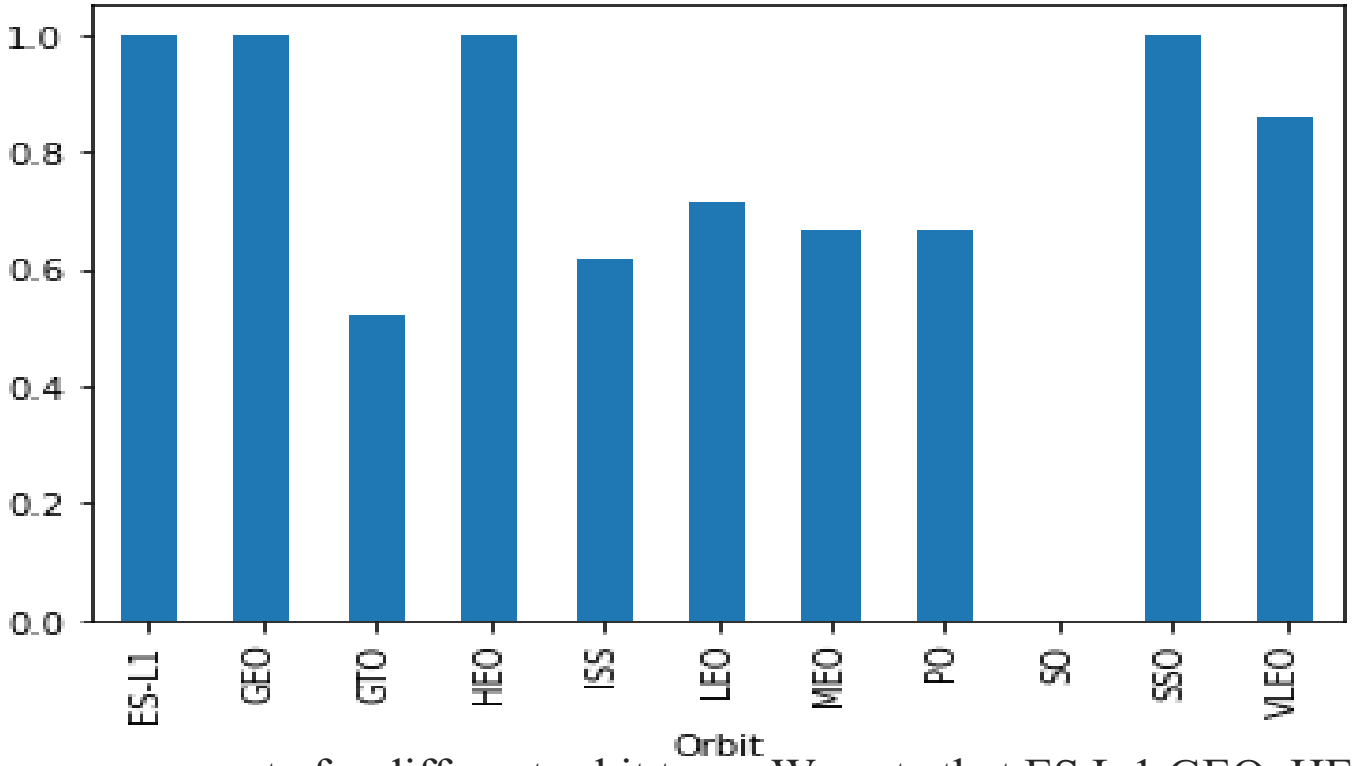


Payload vs Launch Site



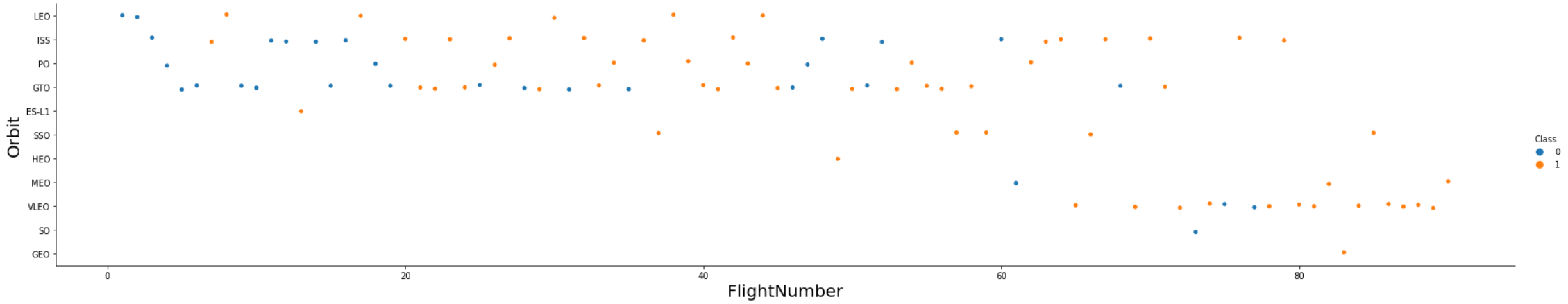
The choice of launch site plays a crucial role in determining whether a heavier payload will lead to a successful landing Conversely, an excessively heavy payload can increase the risk of a failed landing

Success Rate vs. Orbit Type



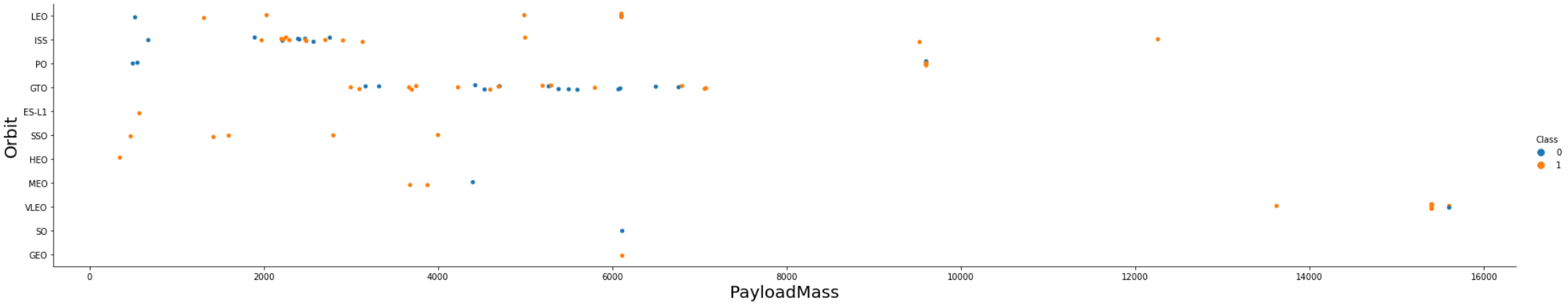
With this plot, we can see success rate for different orbit types We note that ES L 1 GEO, HEO, SSO have the best success rate.

Flight Number vs. Orbit Type



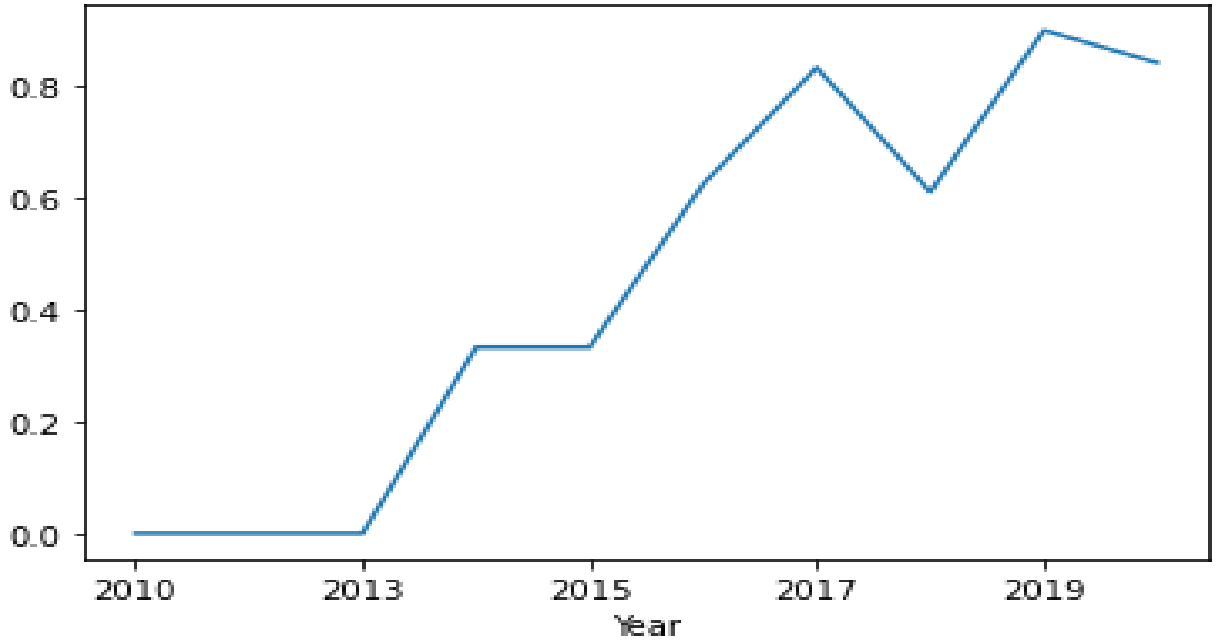
While the success rate tends to increase with the number of flights for Low Earth Orbit (missions, there appears to be no clear correlation between the success rate and the number of Flights for missions to Geostationary Transfer Orbit (However, it's plausible that the high success rates observed in orbits like Sun Synchronous Orbit (or Highly Elliptical Orbit (are influenced by knowledge gained from prior launches in different orbit types

Payload vs OrbitType



The success rate of launches in specific orbits is significantly affected by the payload weight. Notably, heavier payloads tend to enhance the success rate for Low Earth Orbit (LEO) missions. Conversely, reducing the payload weight for launches to Geostationary Transfer Orbit (GTO) has a positive impact on launch success

Launch Success Yearly Trend



All Launch Site Names

SQL Query

```
SELECT DISTINCT "LAUNCH_SITE" FROM SPACEXTBL
```

Results

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Explanation

The DISTINCT Keyword in the query provides the unique values of the attribute "LAUNCH_SITE"

Launch Site Names Begin with 'CCA'

SQL Query

```
SELECT * FROM SPACEXTBL WHERE "LAUNCH_SITE" LIKE '%CCA%' LIMIT 5
```

Results

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer
04-06-2010	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX
08-12-2010	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO
22-05-2012	07:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)
08-10-2012	00:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)
01-03-2013	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)

Explanation

The WHERE clause filters out the LAUNCH_SITES which have the substring “ and we LIMIT’s to show the first 5 records only

Total Payload Mass

SQL Query

```
SELECT SUM("PAYLOAD_MASS__KG_") FROM SPACEXTBL WHERE "CUSTOMER" = 'NASA (CRS)'
```

Results

SUM("PAYLOAD_MASS__KG_")
45596

Explanation

This Query returns the total payload carried by boosters from NASA

Average Payload Mass by F9 v1.1

SQL Query

```
SELECT AVG("PAYLOAD_MASS__KG_") FROM SPACEXTBL WHERE "BOOSTER_VERSION" LIKE '%F9 v1.1%'
```

Results

AVG("PAYLOAD_MASS__KG_")
2534.6666666666665

Explanation

This query returns the average payload mass carried by booster version F9 v1.

First Successful Ground Landing Date

SQL Query

```
SELECT MIN(DATE) AS FIRST_SUCCESS_GP FROM SPACEXTBL WHERE LANDING_OUTCOME = 'Success (ground pad)';
```

Results

FIRST_SUCCESS_GP
2015-12-22

This query returns the dates of the first successful landing outcome on ground pad.

Successful Drone Ship Landing with Payload between 4000 and 6000

SQL Query

```
%sql SELECT "BOOSTER_VERSION" FROM SPACEXTBL WHERE "LANDING _OUTCOME" = 'Success (drone ship)' \
AND "PAYLOAD_MASS__KG_" > 4000 AND "PAYLOAD_MASS__KG_" < 6000;
```

Results

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Explanation

Above query returns the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

Total Number of Successful and Failure Mission Outcomes

SQL Query

```
SELECT MISSION_OUTCOME, COUNT(*) AS QTY FROM SPACEXTBL GROUP BY MISSION_OUTCOME ORDER BY MISSION_OUTCOME;
```

Results

Mission_Outcome	QTY
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

This query returns the total number of successful and failure mission outcomes

Boosters Carried Maximum Payload

SQL Query

```
%sql SELECT DISTINCT "BOOSTER_VERSION" FROM SPACEXTBL \
WHERE "PAYLOAD_MASS__KG_" = (SELECT max("PAYLOAD_MASS__KG_") FROM SPACEXTBL)
```

Explanation

Results

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

We employed a subquery to filter data, utilizing the MAX function to retrieve the highest payload mass. The primary query then utilizes the results from the subquery to identify distinct booster versions (SELECT DISTINCT) associated with the heaviest payload mass.

2015 Launch Records

SQL Query

```
%sql SELECT substr("DATE", 4, 2) AS MONTH, "BOOSTER_VERSION", "LAUNCH_SITE" FROM SPACEXTBL\
WHERE "LANDING _OUTCOME" = 'Failure (drone ship)' and substr("DATE",7,4) = '2015'
```

Results

MONTH	Booster_Version	Launch_Site
01	F9 v1.1 B1012	CCAFS LC-40
04	F9 v1.1 B1015	CCAFS LC-40

This query retrieves the month, booster version, and launch site for unsuccessful landings that occurred in 2015. It processes the date using the Substr function to extract either the month or year. Substr (DATE, 4, 2) is used to extract the month, and Substr (DATE, 7, 4) extracts the

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

SQL Query

```
SELECT LANDING_OUTCOME, COUNT(*) AS QTY FROM SPACEXTBL WHERE DATE BETWEEN '2010-06-04' AND '2017-03-20' \
GROUP BY LANDING_OUTCOME ORDER BY QTY DESC;
```

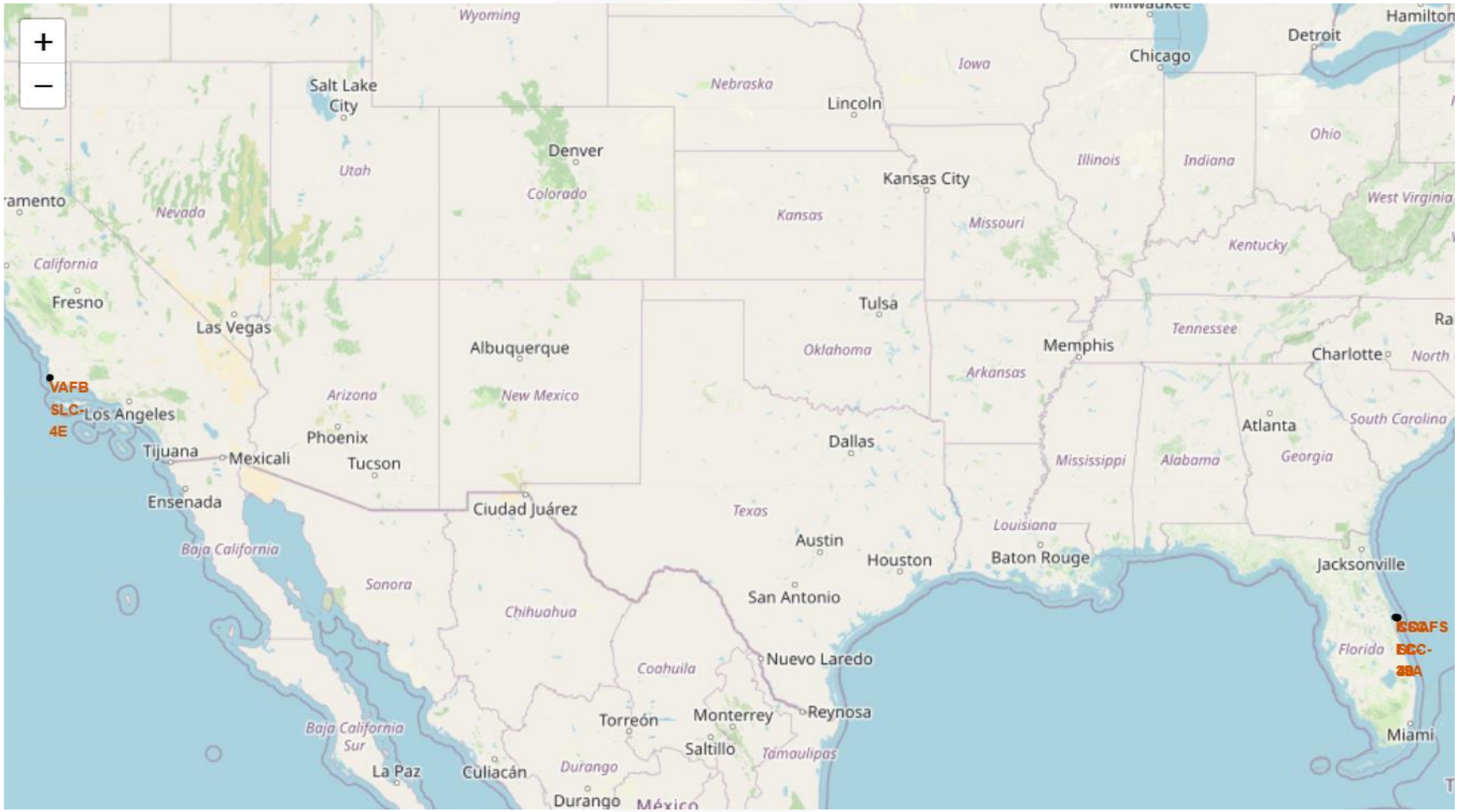
Results

Landing_Outcome	QTY
No attempt	10
Success (ground pad)	5
Success (drone ship)	5
Failure (drone ship)	5
Controlled (ocean)	3
Uncontrolled (ocean)	2
Precluded (drone ship)	1
Failure (parachute)	1

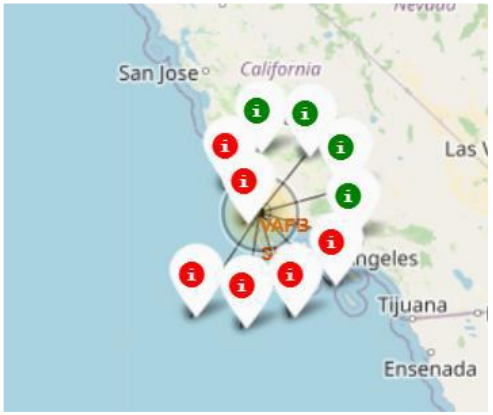
This query ranks the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010 06 04 and 2017 03 20, in descending order.

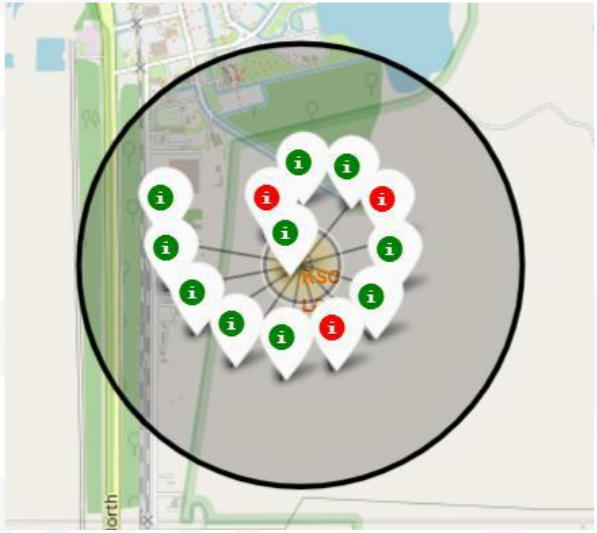
Launch Sites Proximities Analysis

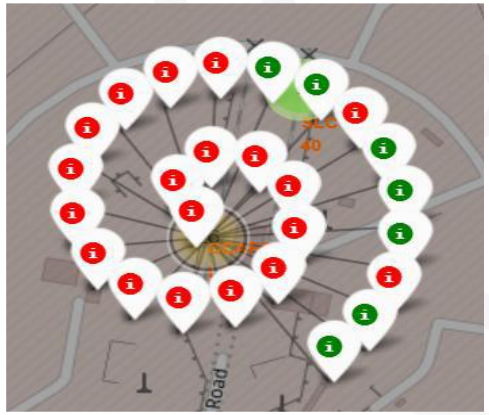
Launch Sites Location

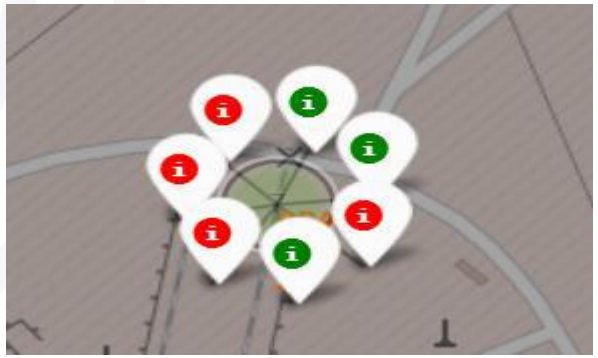


Launch Sites And their Success rate



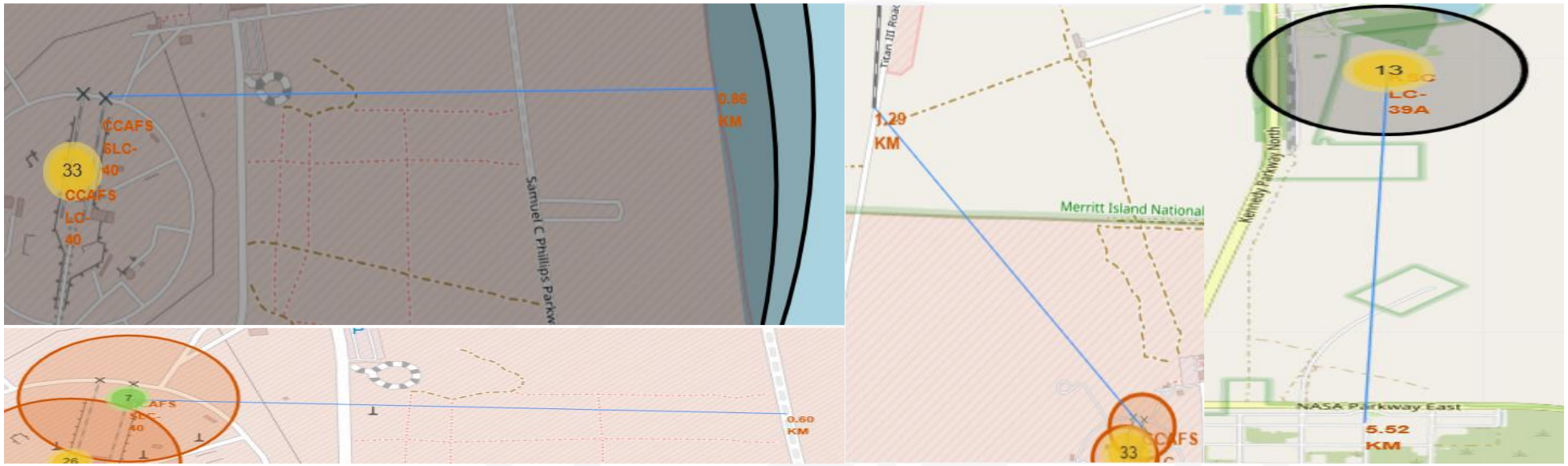






Green marker represents successful launches. Red marker represents unsuccessful launches. We note that KSC LC 39A has a higher launch success rate.

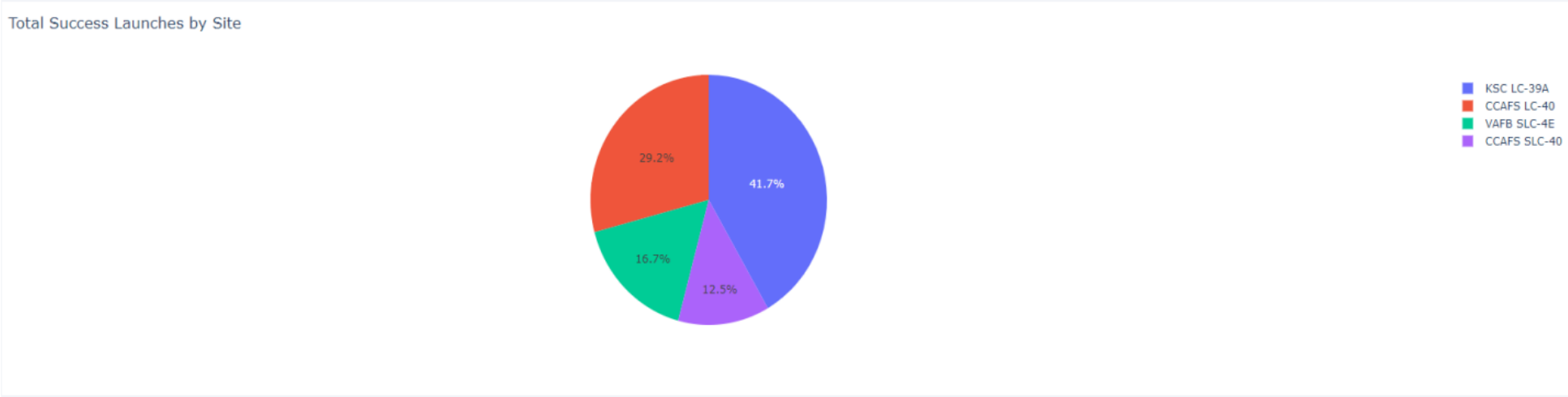
Launch sites and its Proximities



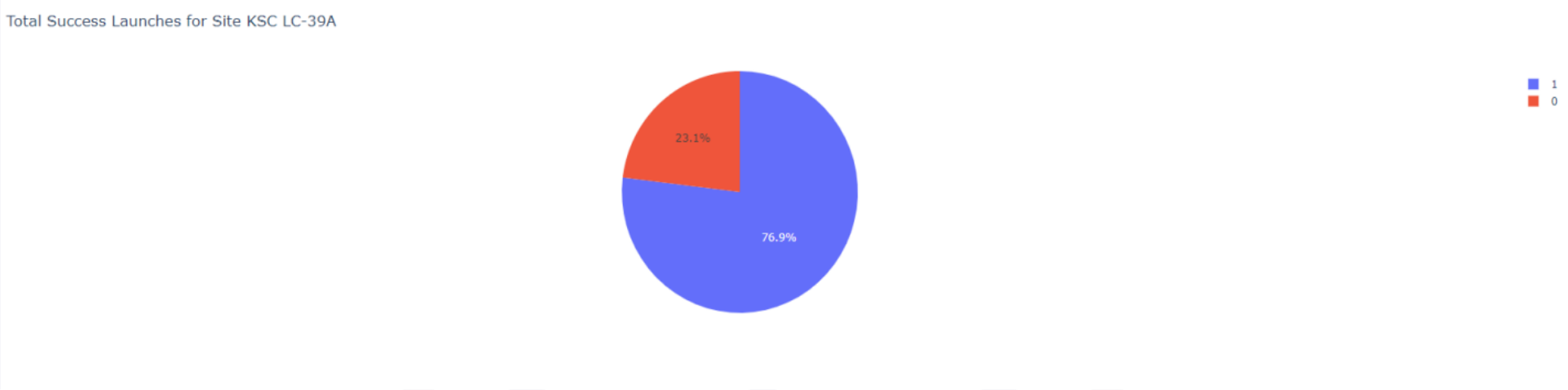
- Are launch sites in close proximity to railways? Yes
- Are launch sites in close proximity to highways? Yes
- Are launch sites in close proximity to coastline? Yes
- Do launch sites keep certain distance away from cities? Yes except CCAFS SLC 40

Build a Dashboard with Plotly Dash

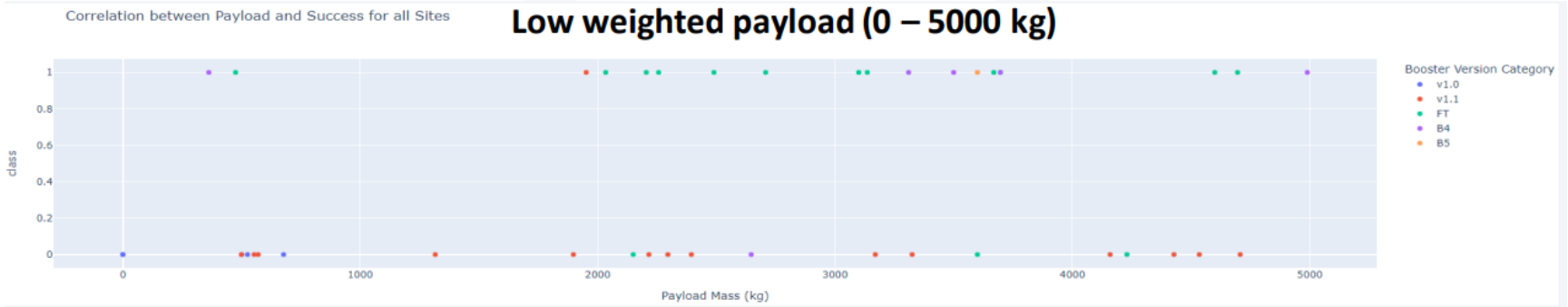
Dashboard –Total Success By Site

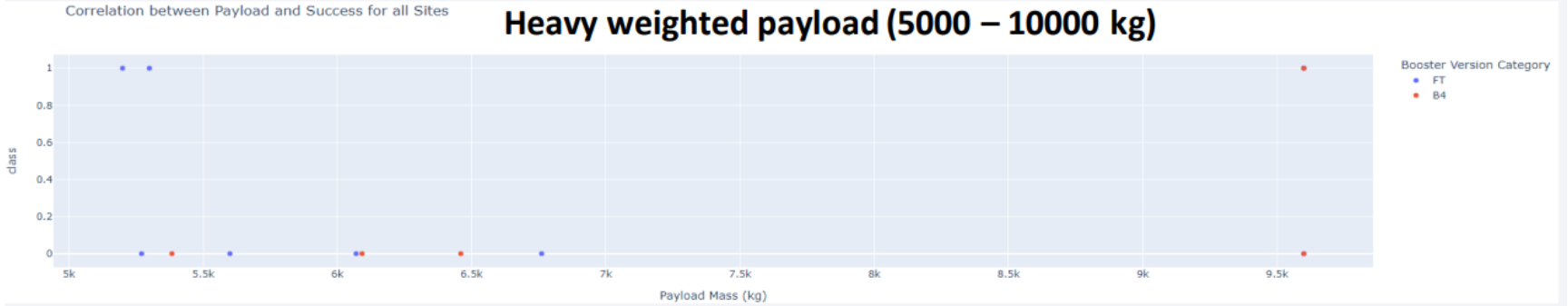


Dashboard –Total Success Launches For KSC LC-39A



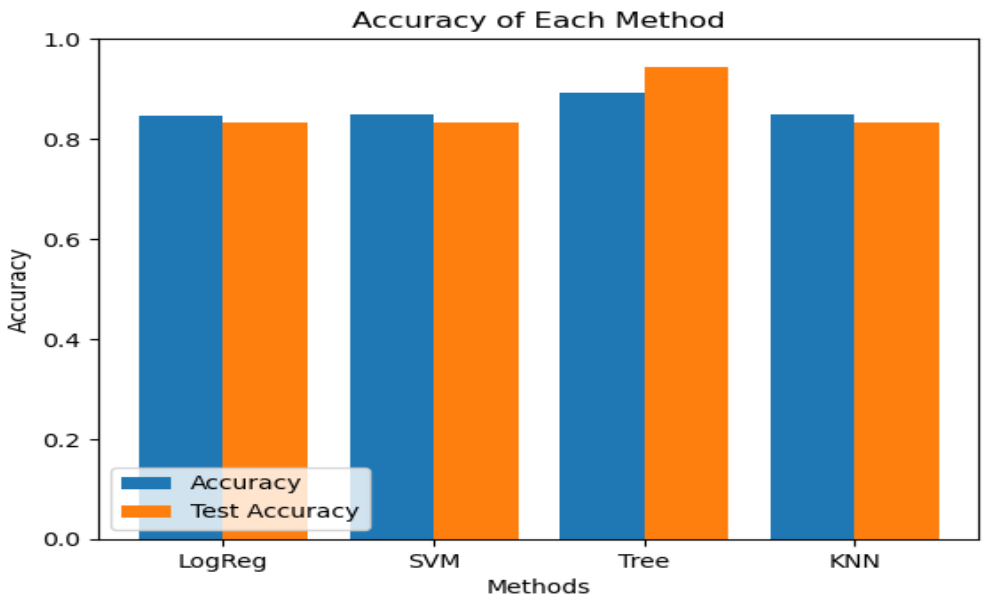
Dashboard –Payload vs Outcome with different Mass





Predictive Analysis (Classification)

Classification Accuracy

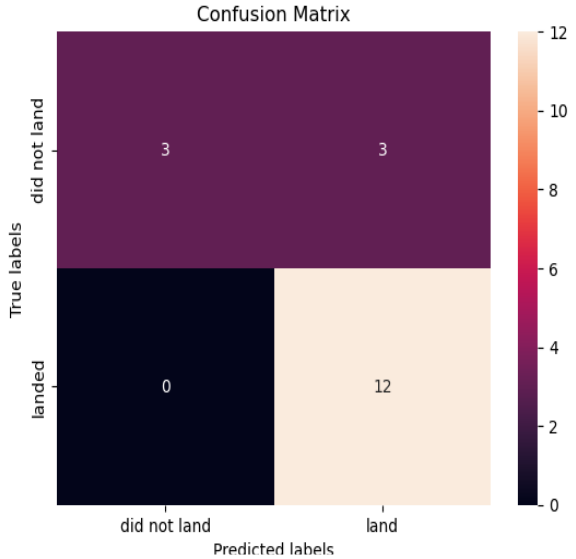


Model	Accuracy	TestAccuracy
LogReg	0.84643	0.83333
SVM	0.84821	0.83333
Tree	0.89107	0.94444
KNN	0.84821	0.83333

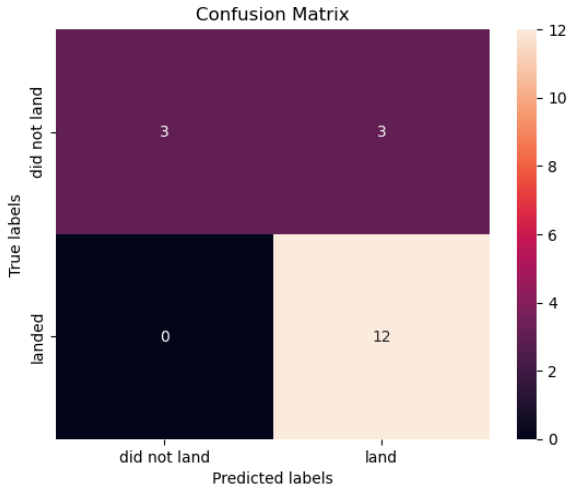
For accuracy test, all methods performed similar. We could get more test data to decide between them. But if we really need to choose one right now, we would take the Decision Tree.

Confusion Matrix

Logistic regression

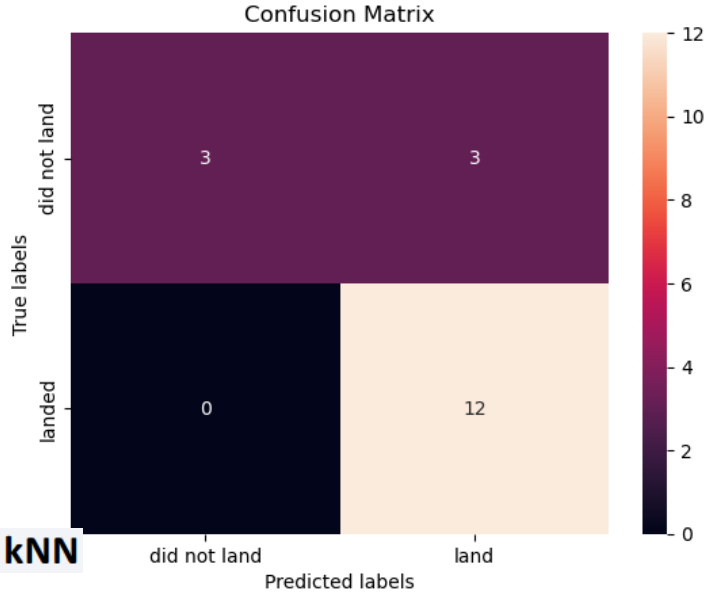


SVM

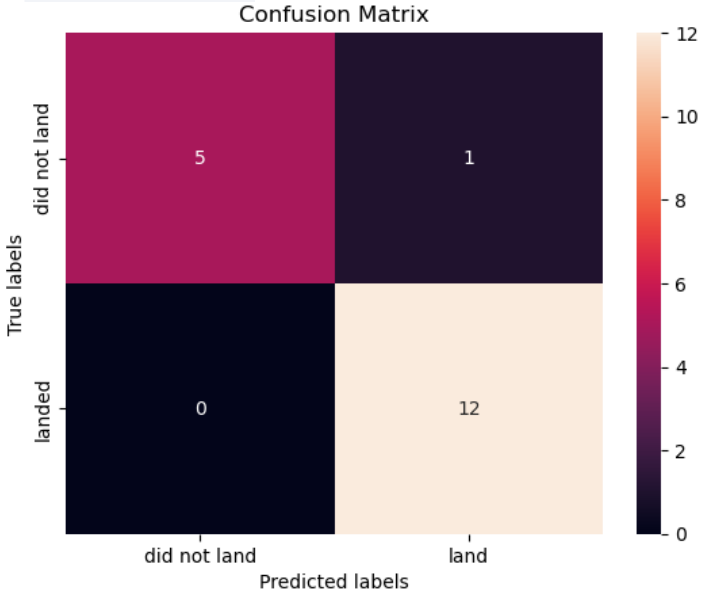


As the test accuracy are almost equal except Decision Tree . The confusion matrices are also identical for others. The main problem of these models are false positives.

kNN



Decision Tree



		Actual values	
		1	0
Predicted values	1	TP	FP
	0	FN	TN

Conclusions

- Launch success depends on factors like launch site, orbit type, payload, and flight number
- Higher success rates observed in orbits like GEO, HEO, SSO, and ES-L1
- Payload mass plays a crucial role in determining landing success
- Success rate improves over time due to experience from previous launches
- Decision Tree model selected as the best model based on performance

Appendix

- **GitHub Repository:**
  - Contains complete project code and notebooks
- **Dataset Source:**
  - Data collected from SpaceX API and Wikipedia
- **Python Code & Notebooks:**
  - Includes data collection, analysis, and modeling steps

Thank You